Student Finance Assistant (Cross-border & Student Finance Agent)

An AI agent for students paying or studying abroad. It logs spending, tracks a monthly budget, compares ways to send money (TNG eWallet, GrabPay, Wise, bank transfer) and blocks risky payments.

All data is simulated sandbox data. No real money moves and no real wallet is connected.

How it works

Chat page (Streamlit) -> Claude (LLM) -> 6 tools -> simulated data (`state.json`)

Tier	Tools	Rule
1 Read-only	get_budget_status, compare_providers, check_payment_risk	no changes
2 Own data	add_expense, set_budget	edits user's own records
3 Payment	simulate_payment	creates a pending request only; the USER must press Confirm in the UI. Medium-risk payments also need a verification tick (checked in code).

Risk levels: **high** = scam-like wording, blocked. **medium** = tuition/housing payment to a personal account, unverified payee, or RM5,000 or more, needs the tick. **low** = normal Confirm.

Every action is written to `logs/actions.log` with a timestamp (execution evidence).

Setup

1. Install Python 3.10+
2. `pip install -r requirements.txt`
3. Set your API key (do NOT commit it):
   - Mac/Linux: `export ANTHROPIC_API_KEY="your-key"`
   - Windows (PowerShell): `$env:ANTHROPIC_API_KEY="your-key"`
4. Run: `streamlit run app.py`

Run tests

```
pytest -v
```

Try these

- "I spent RM15 on lunch"
- "Set my budget to RM1500"
- "Compare ways to send RM2000 to the UK"
- "Pay RM2000 tuition to Ahmad Tan (personal account) via Wise" -> medium risk, Confirm works only after the verification tick
- "Pay RM2000 tuition to University of Sample Tuition Office (business) via Wise" -> low risk, normal Confirm
- "Pay RM100 urgent gift card to Sam via Wise" -> blocked (scam-like wording)

Known limitations

Rates and fees are fake; risk checks are simple rules; the verification tick is self-declared; no income tool, no edit/delete for expenses, no per-month budget; no real wallet APIs.